

Applications of Novel Deep Learning Algorithms for Analysing of Cryptocurrency

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Abstract: With the worldwide spread of cryptocurrencies, data has become richer and more heterogeneous than before and, as a result, poses a new problem related to prediction and decision making. Although there has been recent increase in acceptance of machine learning (ML) and deep learning (DL) technologies, the current analytical models appear to have limited correspondence with cryptocurrency markets. This work directly addresses price forecasting, fraud detection, sentiment analysis, and risk management, thus illustrating the applicability of the current technologies. We implement complex and novel deep learning techniques such as long short-term memory, gated recurrent unit, Bidirectional-LSTM to explore the temporal and spatial behavior of data associated with cryptocurrencies. These models enable accurate predictions of price and market trends along with reinforcement strategies to refine trading strategies and generative models to assess the market and project what the future might require. This is used to help investors and traders spot patterns in the buying and selling of various crypto currencies, these models might have far-reaching effects on the economy. We compared the suggested model's output with that of current setups. This works findings reveal that the proposed model is more accurate than the alternatives since its prediction errors are lower.

Keywords: Cryptocurrencies, price prediction, fraud detection, risk management, LSTM, GRU and Bi-LSTM.

I. INTRODUCTION

Cryptocurrency is a new kind of money that moves across a computer network that is decentralized and not controlled by any central authority [i.e., it is managed not by banks, governments, or corporations]. Asians have been using the internet for decades, and interest and popularity of this technology has increased dramatically and amongst the lot of them many of the dealers and financiers have shifted their focus to cryptocurrency markets including Bitcoin [1]. Data on cryptocurrency is traded in a similar manner to information on stocks, but are on a very simple level heterogeneity. In the first

place, the cryptocurrency markets are open every hour, i.e. Moreover, there are no intermediaries enrolled in cryptocurrency exchanges, and thus, the exchange fees look to be significantly lower. Finally, cryptocurrency markets are defined by their enormous volatility and profound shifts [2]. Also for all these reasons, cryptocurrency markets offer incredible money-earning opportunities to traders, but also represent greater risk.

With money related markets data now becoming available freely and rapidly more than ever before. Therefore, different skilled financial analysts and brokers, they use professional investigation (a tool which is attached to historical market data and gets the dealer the ability to forecast market trends [3]. Specialized investigation provides specialized clues, which are pattern-based attributes of the energy, instability, and drift of an asset. However, when specialized indicators are used, they tend to generate misleading spectral signals so speculators and financial researchers typically use a combination of specialized indicators. Some researchers and financial gamblers used a combination of technical analysis and machine learning methods to predict emerging trends in the stock and cryptocurrency markets [4].

Financial advertising is one of the most complex structures with complexity that has not been validated by the academic literature, and consequently no information on how the components of complex structures are related has been explored. Illustrative examples of the complex nature of the framework are analog exercises in the sense that the architecture of this framework is built on an incremental approach, just as is each subsystem that is aggregated [5]. Unfortunately, in the process of portfolio development, there is an inherent problem in terms of the importance of the information network between different tiers of structure.

The global economy and installment strategies have substantially changed for the last decade mostly because of the

advancement of sophisticated monetary assets enabled by progress in data innovation (IT). These sophisticated monetary instruments, e.g., Bitcoin and Ethereum, have gained the acceptance of clients because of their effectiveness to acquire and grow a client base, and they are backed by blockchain technologies. The blockchain provides a decentralized monetary systems that offer a high level of autonomy, but also poses many problems for governments and financial literacy [6]. At present, Bitcoin and Ethereum are the most widely traded cryptocurrencies with high trading amounts and market capitalization. Faced with the possibility of financial risk for researchers, academics are currently modelling the future value of cryptocurrencies and markets. The below Fig 1 shows the general elements of bitcoin contains blockchain, timestamp, wallets and mining.

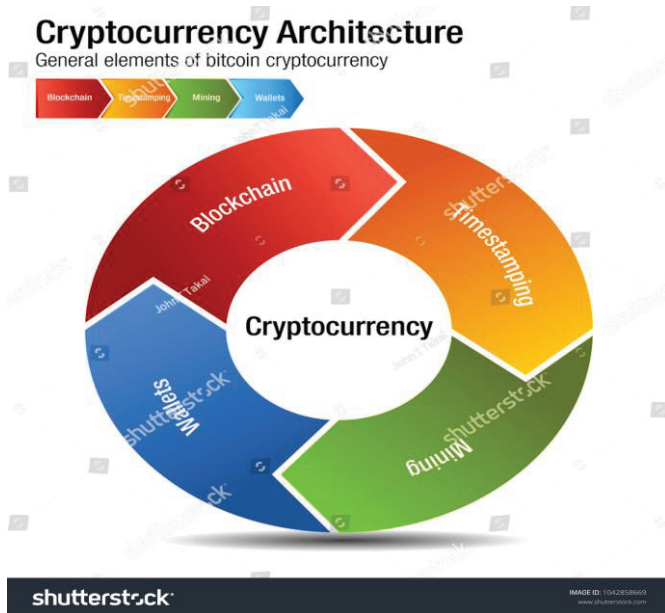


Fig. 1. Cryptocurrency Bitcoin Architecture Chart

Machine learning (ML) algorithms are commonly deployed by numerous contributing companies to decompose the design of information and to read out its key information. These calculations enable the decision maker to identify some nonlinear information designs which other simple computations do not reveal. The calculation of each ML operation changes according to the admissible parameters used and the characteristics of the data set. The speed and accuracy of how a few machine learning operations can break apart huge quantities of historical data is unmatched. Distinctive calculations produce different outcomes on different datasets [7]. The level of calculation accuracy also depends on whether the data is structured or not structured. There are several instances in which the decisionmaker must adapt and rework the dataset to meet the user's unique demands and priorities.

Thus from the old times to the present day, people have used banks and financial institutions for transacting money, and it is money that they interchange. In the current accounting framework people are not in a position to know how much cash is in circulation, or the flow of budgets between all of the customers' accounts, and administrative bodies have access to

such data [8]. The need for accessibility to clients within such a centralized model lends itself to senior managers embezzling and laundering money, an act that is likely to take significant time, if not be discoverable. Apart from this issue, the centralized storage of data by banks and financial institutions forms the basis of every cyber-attack to extract data or capital with the aim to make money for attackers. In addition, there is a need for people to maintain a private personal identity while enabling secure transactions and financial transparency [9].

Bitcoin (BTC) is the first credible digitized digital money to take root on blockchain technology, invented by an unidentifiable author or a collective (Satoshi Nakamoto). In Bitcoin and related cryptocurrencies, the cryptographic address serves as identity of individuals, and, because the addresses are scrambled, both the identity of the users and the exchanging intent are blurred [10]. The anonymity and pseudonymity of the individual's identity during transaction and the huge amount of users' accounts and the increase in the number of clients have attracted a large number of illegal activities to this network, such as money laundering, cheating, and even financing criminal activities such as psychological warfare.

There are few cryptocurrencies other than Bitcoin today that are widely used for commercial transactions, and enabling currency conversion is one of the most important and common center capabilities. All of these money standards offer new facilities to clients and they aim to be top of their class with providing increased security and speed of transactions. Engineers of unusual coins make every effort to find new and unusual characteristics absent from others, refine them, and to offer a claim structure. Anti-Money Washing (AML) solutions have been implemented by cryptocurrency insight firms in a number of ways via the tracking of the transfer of money forms and the use of machine learning techniques with, along with, other methods such as the quantitative and graph-based approach [11].

The think eventually looks to upgrade the understanding of how social media input impacts showcase conduct, advertising commonsense suggestions for financial specialists and investigators pointing to explore the unstable scene of cryptocurrency markets.

II. LITERATURE SURVEY

A significant amount of research has used various ML and DL algorithms to forecast Bitcoin values and isolate key variables. Ahmad EI Majzoub [12], it tests several machine learning and deep learning applications for cryptocurrency profit prediction. Their study explores different algorithms and their interpretability features rather than detailing a specific proposed system. The advantages are predicting cryptocurrency returns is possible with various algorithms while multiple algorithms offer different features and limitations. The disadvantage is prediction algorithms may not generalise across different assets.

MA Mohammed [13], these are focused on assessing the efficacy of blockchain technology in financial accounting and its implications for the financial sector. The study highlights the transformative potential of blockchain and suggests directions for future technological advancements and

application areas. The advantages are enhanced efficiency and accuracy in financial accounting. The disadvantages are elevated transaction fees hinder broader adoption and technical constraints pose significant challenges. V Kochliaridis [14], the proposed system, Trader Net-CR, is a risk-sensitive cryptocurrency-trading system consisting of different modules. This hybrid approach aims for the best possible net profit and loss (PNL) returns under the minimum risk of trading and maximized portfolio performance indicators in total. The advantage combines DRL with technical analysis for better trading strategies and maximizes profit while minimizing trading risks effectively. The disadvantages obtain DRL techniques with lack of risk-awareness in trading strategies and difficulty in maximizing PNL while minimizing trading risks.

Z Shahbazi [15], the current method is designed for exploring professional accounting in the context of the risk associated with bitcoin and the foreseen effect on the financial accounts. The system analyses the highest likelihood risks associated with determined cryptocurrencies. It employs Reinforcement Learning (RL) techniques for risk management and asset allocation within the cryptocurrency network. The advantages include better risk management through Hierarchical Risk Parity and performing robust models for various re-balancing intervals. The disadvantage is the high likelihood of private key loss risks funds and anonymity complicates transaction identification and accountability.

The below Fig 2 provides a survey of the different areas of study related to Bitcoin, such as blocks, blockchains, the mining procedure, and proof of work.

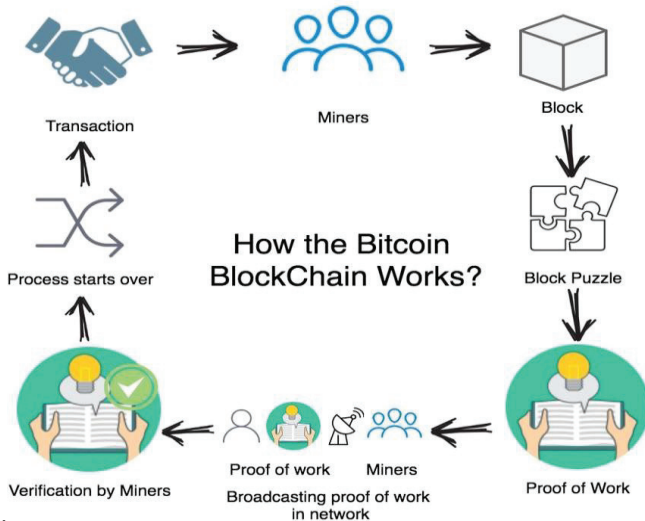


Fig. 2. Bitcoin Cryptocurrency and Mining

S Chen [16], the proposed approach forecasts the value of Bitcoin by a modified deep neural network (DNN) algorithm in order to remove the financial risk regarding electronic commercial transactions using digital currency. The system collects data from new addresses including net flow over accounts and transaction volume. The advantages include improved prediction accuracy for Bitcoin price forecasting and reduced financial risks in digital currency transactions. The disadvantages may be that Standard machine learning

techniques may not work effectively and also digital currency involves significant financial risks. technical constraints pose significant challenges.

A Heidari and B Bahrak [17], the proposed approach classifies the Bitcoin transaction using graph convolutional network based on deep learning model in combination with multi-layer perceptron. It extracts several features from the transaction graph and performs preprocessing on the data. The advantage utilizes graph-based features to improve model performance. The disadvantages are an imbalanced dataset with many unknown transaction labels and previous methods lacked the power to detect all illegal transactions.

J Park and YS Seo [18], the system has features, which can be used as indices for decision making process so as to assist the clients in optimizing their profit maximization strategy. It utilizes three input features they are sell profit, buy profit, and max profit. The advantages are provided by improving action recommendations for profit maximization. The disadvantages include the low performance of existing action models and clients facing complex decision-making for profits.

S Tanwar [19], the system is suggested deep-learning hybrid model that blends GRU and LSTM networks. The model is capable of real-time price prediction and was trained and tested on benchmark datasets. The advantages available are real-time application for investors and high accuracy in price forecasting compared to existing models. The disadvantage is the struggle to differentiate itself from Bitcoin A Sankaya and S Aslan [20], it is based on deep learning architectures and machine learning techniques to study the Bitcoin sentiment and its resulting effect on coin's price. The study utilizes a text blob for initial sentiment assessment and employs basic machine learning algorithms alongside CNN, LSTM, and deep learning architectures for sentiment classification. The advantages contained better performance with fewer data for text classification. The disadvantages are high noise levels in Twitter data complicate analysis and unstructured data requires extensive pre-processing for accuracy.

III. METHODOLOGY

This section involves for analyzing cryptocurrency price prediction using a hybrid approach by these algorithms LSTM, GRU, and BI-LSTM.

A. Long Short-Term Memory (LSTM):

The input gate, forget gate, and output gate, control the amount of data sequence needed for the network to make a prediction as shown in Fig 3.

- Input gate: After data importation, information first passes through an input gate. The on/off switch decides whether the information is stored or not, depending on the state of the cell.
- Output gate: It determines the amount of output data.
- Forget gate: It makes the decision whether to retain or discard the information.

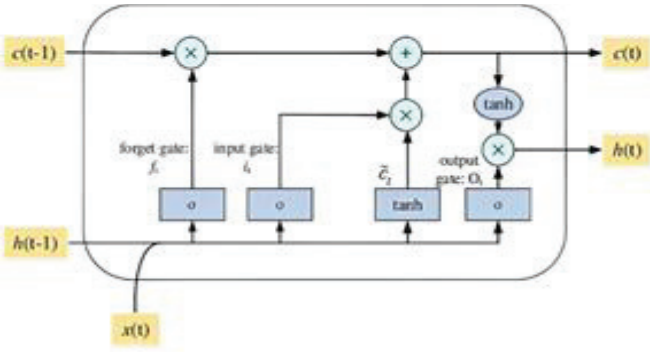


Fig. 3. Schematic diagram of LSTM

B. Gated Recurrent Unit (GRU):

The gate for updating and resetting. GRUs are effective against the vanishing gradient problem in RNNs and, at the same time, improve the structure of the LSTM model. Both gates may archive useful information in the memory cell and transmit information to the next stage of the network. Performance-wise, GRU and LSTM show no difference when comparing different test environments. Fig 4 displays the organization of GRU units.

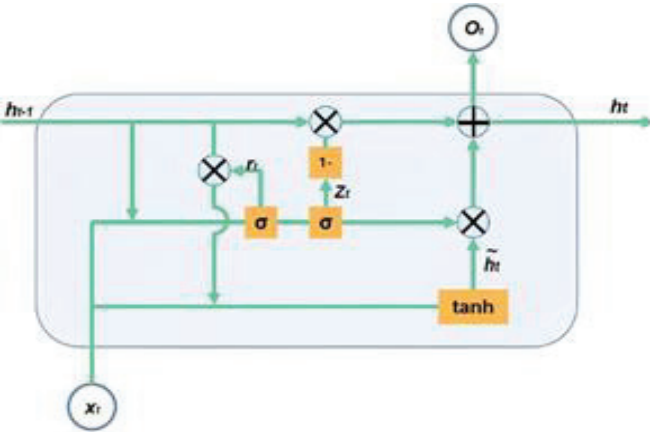


Fig. 4. GRU unit structure

C. Bidirectional Long Short-Term Memory (Bi-LSTM):

The Bi-LSTM model can learn context information from feature sequences and incorporate forward and backward

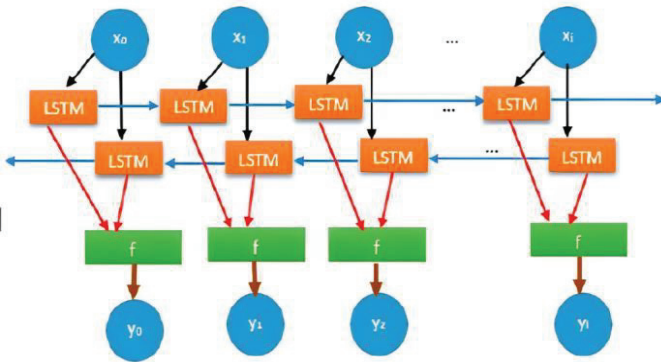


Fig. 5. Bi-LSTM Structure

dependencies. Bi-LSTM allows to predict ahead by fusing a forward LSTM which processes the sequence in temporal order with a backward LSTM which processes the series in backward order. The output is finally synthesized by appending the forward and reverse states of the LSTM, as shown in fig5.

D. Applicability Of Hybrid Approach For Crypto Currency Price Prediction

Due to its distinct capability to integrate the strengths of different deep learning models it is very suitable to use for the hybrid approach with regard to the cryptocurrency price prediction. This method solves the problems of the fluctuations and irreversibility of cryptocurrency markets. Combining LSTM, GRU, and Bi-LSTM, the hybrid model extracts sequence patterns, long-term dependencies, and bidirectional patterns from the time-series, in order to provide a holistic interpretation of price history.

It exploits LSTM for sequential data learning, GRU for efficient learning of long-range dependencies while having lower computational complexity, and Bi-LSTM to learn the forward and reverse time-series dependencies. Moreover, the feature selection works through statistical techniques to choose those features which are significant e.g., "Close" price, and those capabilities are increased through normalization techniques such as Min Max Scaler.

The robustness of the hybrid model is also improved by ensemble methods (e.g., stacking or weighted averaging) which merge together the outputs of the models themselves to better achieve overall prediction accuracy. By giving much better accuracy, and interpretability of results, the hybrid method is perfectly suited for prediction of future cryptocurrency prices to assist more practically with better investment and trading choices. This owing to its great intelligence and flexible nature furnishes a strong solution for tackling the challenges in cryptocurrency price prediction.

E. ALGORITHM

Input: Bitcoin Historical Data

Output: Predict future cryptocurrency prices based on past trends and use the results to make better investment or trading decisions.

1. Start
2. Consider the Bitcoin Historical Data Dataset (Data Collection)
3. Perform Data Preprocessing for converting Unix Timestamp to Date Time
4. Data Cleaning for handling missing values
5. Feature Scaling is to normalize the data for selected price-related columns to bring value b/n 0 and 1
6. Build an efficient Hybrid Model using LSTM, Bi-LSTM, and GRU algorithms.

At first Apply LSTM to learn sequential data patterns

Next, Use GRU to handle long-term dependencies efficiently

Finally apply Bi-LSTM for capturing both forward and backward dependencies in a sequence from the dataset

7. Evaluate the Model Performance in terms of MSE, MAE and R2 score

8. Visualize the Predicted Results

Plot the actual vs. predicted cryptocurrency prices for each model.

Compare the performance of the three models

9. Stop

Fig 6 illustrates that the workflow begins by acquiring Bitcoin Historical Data consisting of meaningful columns such as Timestamp, Open, High, Low, Close and Volume. Data preprocessing includes transforming the Unix Timestamp to a human-readable date-time, to re-use data in time-series applications. Imputation of missing values is carried out during data preprocessing to get a homogeneous dataset. Feature selection is based on the 'Close' price as the primary predictor variable, dropping unnecessary features. A hybrid model is designed by integrating LSTM to capture sequential pattern of data, GRU to make effective use of the long-term dependence and Bi-LSTM to represent the forward and backward dependence of data. Models are then assessed based on Mean Squared Error (MSE), Mean Absolute Error (MAE), and R^2 score before plotting actual versus predicted cryptocurrency values and model performance comparisons.

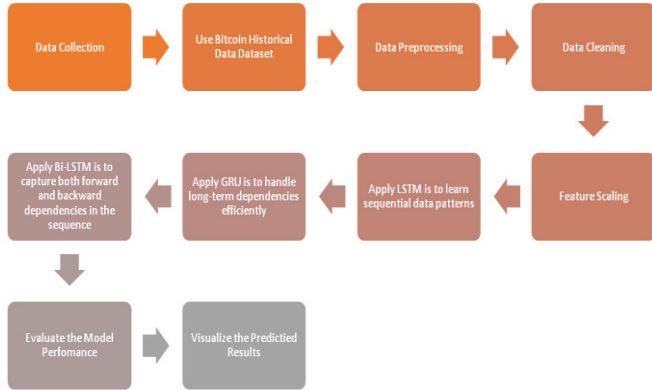


Fig. 6. Step-by-Step proposed methodology

F. Dataset Description

The dataset we used for this work is Bitcoin Historical Data at 1-minute intervals from selected exchanges observed over a period, from 2012 to now. The main columns of the dataset generally comprise the Timestamp which should record the precise moment at which each entry occurred in Unix format, and close, open, High, and Low (OHLC) price for each minute of trading. Also, to the dataset, the Volume that corresponds to the trading unit (bitcoin) in that minute is included. This dataset contains a valuable time-series dataset of the fluctuations in Bitcoin prices that can be used to help predict and model the market's behavior of Bitcoin. The Timestamp is crucial in the process of building time-series features and

guaranteeing that the data is well chronological order to fit in the prediction tasks. Preprocessing operations including Unix timestamp conversion into a date-time format (readily) normalization of features, e.g., Close price, and imputation of missing values are carried out prior to training of deep-learning models.

IV. RESULTS AND DISCUSSION:

We utilized the models to forecast prices for the Bitcoin Historical datasets independently in order to assess their efficacy. The goal of using the suggested deep learning models was to make reliable long-term predictions about the value of different crypto currencies. Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Normalized Root Mean Squared Error (NRMSE) were some of the evaluation metrics used to find prediction mistakes. A portion of the dataset was reserved for testing purposes, while the remaining 70% was used for training purposes to construct the model. Cryptocurrency price predictions were made using the LSTM model. Examining the model's accuracy in comparison to others was the primary motivation for this work.

Various software and hardware environments were utilized in the development of the suggested system. Prior work on deep learning models made use of the Tensor Flow and Keras frameworks. For the normalization phase, Scikit-learn was utilized. This program was set up with a 3.20 GHz Intel (R) Core (TM) i7-4770 processor, 8 GB of RAM, and 64-bit Windows 10. Over the course of 180 days, from August 15, 2023, to September 1, 2023,

The suggested hybrid deep learning models were designed to predict the future values of the Bitcoin historical datasets. As you can see from Figure 7, we have compared many models graphically. On a scale from 0 to 1, Table 1 displays the results of both the proposed and current methods for forecasting.

TABLE I. RESULT OF VARIOUS ALGORITHMS

Parameter	LSTM	GRU	Bi-LSTM	Proposed Approach
MSE	0.0087	0.0090	0.0119	0.0082
MAE	0.0073	0.0078	0.0087	0.0072
R2 Score	0.9771	0.9760	0.9684	0.9784

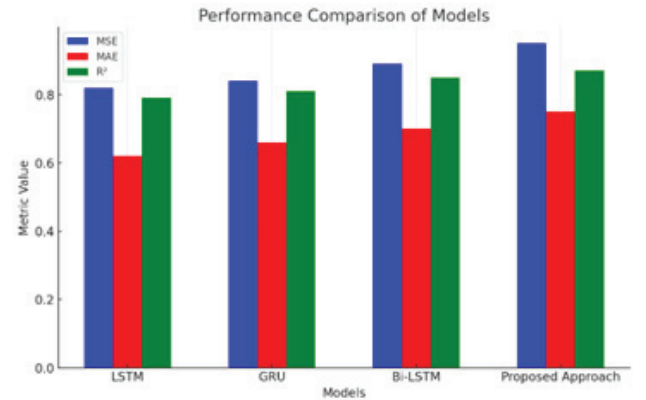


Fig. 7. Comparative Result of the Proposed Approach

The above figure 7 illustrates that cryptocurrency data in the Bitcoin Historical data sets in our environment is assessed in terms of RMSE, MSE, and NRMSE. All the parameters are assessed on a 1 point scale in all the four algorithms RNN, Bi-LSTM, Bi-GRU and our proposed Hybrid approach. The Figure 7 and table 1 clearly shows that our proposed Hybrid approach shows outstanding performance on Bitcoin Historical dataset when compared to all the algorithms which is very significant.

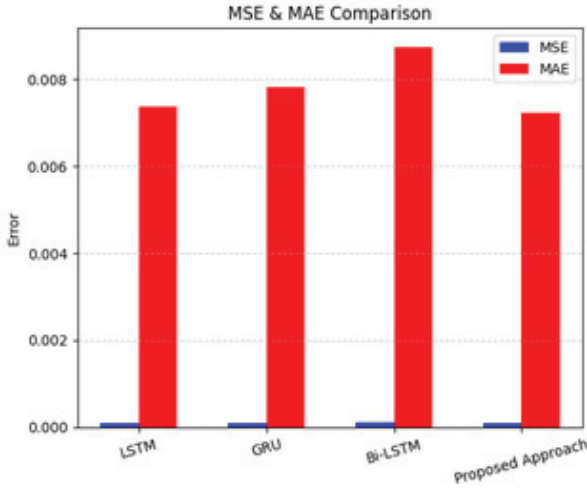


Fig. 8. .Comparative Result of MAE & MSE

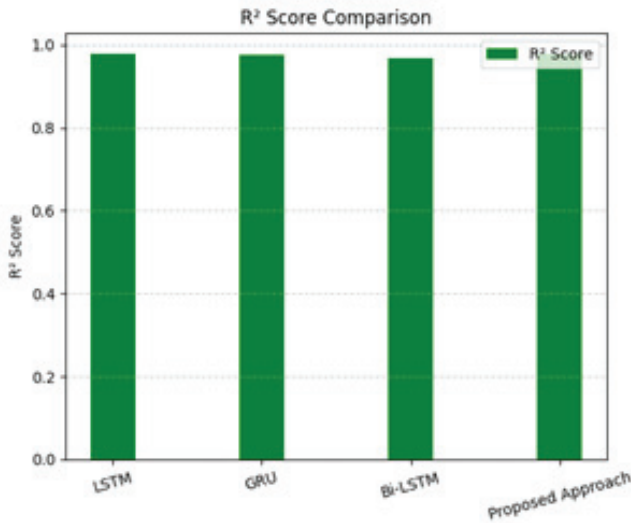


Fig. 9. Comparative Result of R²

This twofold advantage highlights the Hybrid model's efficiency in terms of MAE, MSE & R² predicting cryptocurrency Prices, an important part of market investments.

V. CONCLUSION

In this work, the LSTM, GRU, and Bi-LSTM models are used for the prediction of the price of Bitcoin based on historical data. Data preprocessing, feature selection, and hybrid model training enabled us to identify the underlying patterns and couplings of the market. Models were assessed

with performance measures including MSE, MAE, and R², all of which describe the models capacity to price forecasts in a realistic manner. This paper shows the application of deep learning for cryptocurrency making and thus contributes to the investors and traders in a very volatile market. Finally, To predict cryptocurrency returns, this work employed a novel Deep Learning approach rather than the conventional time series models. We discovered that this novel approach yielded reliable return classification forecasts.

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